MTH301 - Midsem 2

1. Let K be a (covering) compact subset of a metric space (X, d). Show that every open cover of K has a Lebesgue number. Define uniformly continuous function and show that $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x) = x^2$ for each $x \in \mathbb{R}$ is not uniformly continuous. $(2 + 1 + 2 = 5)$

2. Describe types of discontinuities of $f : (a, b) \to \mathbb{R}$ and give examples of discontinuities of each kind. Show that for a monotonically increasing function f and $x \in (a, b)$, the right hand limit $f(x+)$ is given by $f(x+) = \inf\{f(t) : t \in (x, b)\}$. $(2 + 1 + 2 = 5)$

   OR

   State and prove the second fundamental theorem of calculus. If $g : [-1, 1] \to \mathbb{R}$ is defined by $g(x) = \begin{cases} 0 & \text{if } -1 \leq x \leq 0 \\ 1 & \text{if } 0 < x \leq 1 \end{cases}$, then, show that g has no primitive. $(1 + 2 + 2 = 5)$

3. (a) State L'Hôpital's Rule. Determine $\lim\limits_{y \to \infty} \dfrac{y^2}{e^y}$ $(1 + 1 = 2)$

   (b) Let $f : (a, b) \to \mathbb{R}$ be a 2-times differentiable function. Suppose $f''(x) \geq 0$ for all $x \in (a, b)$. Then, show that f is a convex function on (a, b). (3)

4. Define Darboux Integrability. State Riemann's Integrability criterion. Let m be a positive integer ≥ 1000 and $A = \left\{\frac{p}{q} \in [0, 1] \cap \mathbb{Q} : 1 \leq q \leq m\right\}$. Let $f = \chi_A$ be the characteristic function of A given by $f(x) = \begin{cases} 1 & x \in A \\ 0 & x \in [0, 1] \setminus A \end{cases}$. Use Riemann's integrability criterion to show that $f \in \mathcal{R}[0, 1]$. $(2 + 1 + 2 = 5)$